

Name: \_\_\_\_\_



## Generating a sequence using a position-to-term rule

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1 A term-to term rule describes how to calculate the next term from.... Tick **1** correct answer

☐ the constant additive rule

☐ the multiplier

☐ the previous term

☐ the term number

2 True or false. In the sequence 0, 1, 2, 3, 4, ... the 2nd term is 3. Tick **1** correct answer

☐ True- the third number is '2'

☐ False- The second number is not '3'

☐ There is no way of knowing without the term-to-term rule

3 If the 7 times table start with  $1 \times 7 = 7$  what is the 100th number in the 7 times table?  
Fill in the blank

\_\_\_\_\_



4 Here are the first 3 terms in a pictorial sequence. Which of these rules could refer to the number of lines in each term? Tick **2** correct answers



- ☐ add 2 to the previous term to get the next term
- ☐ add 3 to the previous term to get the next term
- ☐ add 5 to the previous term to get the next term
- ☐ there are 3 lines per section plus 2 for the base.
- ☐ there are 5 lines per section

5 Which of these is an advantage for using a term-to-term rule Tick **2** correct answers

- ☐ They are easy to use to find the next term in a sequence
- ☐ They are easy to use to find any term in a sequence
- ☐ They work for all sequences where there is a rule between successive terms
- ☐ They are unique, the same rule will always produce the same sequence.
- ☐ There is only one term-to-term rule that will work if you know the first 3 term.



6 Match the sequences with the term-to-term rules. Write the correct letter in each box

|   |                       |
|---|-----------------------|
| a | -2, -6, -18, -54, ... |
| b | -2, -6, -10, -14, ... |
| c | -2, -6, -14, -30, ... |
| d | -2, -6, -12, -20, ... |
| e | -2, -6, -14, -26, ... |
| f | -2, -6, -8, -8, ...   |

|  |                                                 |
|--|-------------------------------------------------|
|  | multiply the previous term by 2 then subtract 2 |
|  | add - 4 to the previous term                    |
|  | multiply the previous term by 3                 |
|  | add - 4, add - 2, add 0, ...                    |
|  | add - 4, add - 8, add - 12, ...                 |
|  | add - 4, add - 6, add - 8, ...                  |

